

## Proving triangles similar



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To prove:  $\triangle ABC \sim \triangle XYZ$

| Statements                            | Reasons                                  |
|---------------------------------------|------------------------------------------|
| 1. $\frac{AC}{XZ} = \frac{BC}{YZ}$    | Given                                    |
| 2. $\angle ACB \cong \angle XZY$      | Given                                    |
| 3. $\triangle ABC \sim \triangle XYZ$ | Side-angle-side similarity (SAS $\sim$ ) |

2

To prove: The two triangles are similar.

| Statements                                         | Reasons                                 |
|----------------------------------------------------|-----------------------------------------|
| 1. $\frac{AC}{YZ} = \frac{AB}{XZ} = \frac{CB}{XY}$ | Given                                   |
| 2. $\triangle ABC \sim \triangle ZYX$              | Side-side-side similarity (SSS $\sim$ ) |

3

To prove: The two triangles are similar.

| Statements                             | Reasons                             |
|----------------------------------------|-------------------------------------|
| 1. $\angle IHJ \cong \angle LKJ$       | Given                               |
| 2. $\angle IJH \cong \angle LJK$       | Vertical angles congruence theorem  |
| 3. $\triangle HIJ \sim \triangle K LJ$ | Angle-angle similarity (AA $\sim$ ) |

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To prove: The two triangles are similar.

| Statements                            | Reasons                                  |
|---------------------------------------|------------------------------------------|
| 1. $\frac{AC}{BC} = \frac{EC}{DC}$    | Given                                    |
| 2. $\angle ACE \cong \angle BCD$      | Reflexive property of congruence         |
| 3. $\triangle ACE \sim \triangle BCD$ | Side-angle-side similarity (SAS $\sim$ ) |

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To prove: The two triangles are similar.

| Statements                                 | Reasons                             |
|--------------------------------------------|-------------------------------------|
| 1. $\overline{AB} \parallel \overline{DE}$ | Given                               |
| 2. $\angle BAC \cong \angle EDC$           | Alternate interior angles theorem   |
| 3. $\angle DCE \cong \angle ACB$           | Vertical angles congruence theorem  |
| 4. $\triangle ABC \sim \triangle DEC$      | Angle-angle similarity (AA $\sim$ ) |

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To prove:  $\triangle ACE \sim \triangle BCD$

| Statements                            | Reasons                             |
|---------------------------------------|-------------------------------------|
| 1. $\angle CBD \cong \angle CAE$      | Given                               |
| 2. $\angle BCD \cong \angle ACE$      | Reflexive Property of Equality      |
| 3. $\triangle ACE \sim \triangle BCD$ | Angle-angle similarity (AA $\sim$ ) |

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To prove:  $\triangle ABC \sim \triangle DEC$

| Statements                            | Reasons                                  |
|---------------------------------------|------------------------------------------|
| 1. $\frac{BC}{CE} = \frac{AC}{CD}$    | Given                                    |
| 2. $\angle BCA \cong \angle ECD$      | Vertical angles congruence theorem       |
| 3. $\triangle ABC \sim \triangle DEC$ | Side-angle-side similarity (SAS $\sim$ ) |

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To prove:  $\triangle ABC \sim \triangle DEC$

| Statements                            | Reasons                             |
|---------------------------------------|-------------------------------------|
| 1. $\angle BAC \cong \angle EDC$      | Given                               |
| 2. $\angle BCA \cong \angle ECD$      | Vertical angles congruence theorem  |
| 3. $\triangle ABC \sim \triangle DEC$ | Angle-angle similarity (AA $\sim$ ) |



## Additional questions

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- a True
- b False
- c False
- d False
- e True

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- a  $\triangle AYP$  is not similar to  $\triangle BOX$  because  $\angle A \not\cong \angle B$ .
- b  $\triangle JKL \sim \triangle JMN$  by the SAS similarity theorem.
- c  $\triangle UVW \sim \triangle FGH$  by the SAS similarity theorem.
- d  $\triangle ZYG \sim \triangle BRP$  by the SSS similarity theorem.

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- a Since  $\angle A \cong \angle F$  and  $\angle B \cong \angle C$ ,  $\triangle ABC \sim \triangle FCE$  by the AA similarity theorem.
- b Since  $\angle MVL \cong \angle UVT$  and  $\frac{MV}{VT} = \frac{LV}{VU}$ ,  $\triangle MVL \sim \triangle TVU$  by the SAS similarity theorem.
- c Since  $\angle U \cong \angle F$  and  $\frac{UW}{HF} = \frac{UV}{FG}$ ,  $\triangle UVW \sim \triangle FGH$  by the SAS similarity theorem.
- d Since  $\frac{FR}{FE} = \frac{FS}{FD} = \frac{SR}{DE}$ ,  $\triangle FSR \sim \triangle FDE$  by the SSS similarity theorem.

12  $x = 11$